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Basic word order typology revisited: a crosslinguistic quantitative study based on UD and WALS

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Abstract: This study quantitatively examines the first five universals of Greenberg's basic word order typology based on 74 large-scale annotated corpora from two perspectives. The results show that (1) the dominant orders extracted from corpora concur with those retrieved from the *World Atlas of Language Structures* (henceforth, WALS) and provide knowledge of dominant orders to languages absent in the WALS, demonstrating the feasibility of adopting corpora to determine dominant orders in typological studies; (2) approaching word order as a discrete variable suggests that the relative order of adjective and noun cannot be predicted by the relative orders of object and verb and genitive and noun, which means the violation of Greenberg's related universal; (3) approaching word order as a continuous variable also indicates the violation of this universal; and (4) the language samples based on the annotated corpora database further demonstrates that languages that are in line with this universal are rare and internally heterogeneous. Our findings suggest the possibility of drawing typological conclusions based on the frequencies and probabilities extracted from corpora materials and demonstrate that a more cautious adoption of the well-known universals is needed, indicating the importance of viewing word order features from various perspectives to better capture the characteristics of natural languages.

Keywords: annotated corpus; dominant order; probability; Universal Dependencies; word order typology

1 Introduction

The pioneering work of Greenberg (1963) marked the beginning of word order typology. Ever since then, many studies have been conducted on word order universals using extensive language samples (e.g., Bickel 2015; Dryer 1988, 1992, 2005; Hawkins 1983; Siewierska 1998), and many typological databases have been developed (e.g., Bickel et al. 2022; Dryer and Haspelmath 2013). Such typological studies and databases have established the status of many word order universals in linguistic typology and provided the basis for applying word order knowledge in multilingual natural language processing.

Notwithstanding these advances, limitations in linguistic materials of previous studies and databases have also emerged. Most of these studies are largely based on secondary linguistic resources, which might undermine the reliability and veracity of their findings (Alzetta et al. 2019). There is also an increasing recognition that the status of word order universals should be further examined using other materials (Chen and Gerdes 2018; Levshina 2019; Liu 2010).

For these reasons, annotated corpora or treebanks (Abeillé 2003) are beginning to be used as a new resource for investigating word order typology (Gerdes et al. 2021). Empirical results show that corpus-based typological research might better capture the unique characteristics of intra-linguistic variation (Levshina 2019). Although the corpus-based approach has its own set of problems that the field has not yet been able to solve (e.g., differing

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text lengths, unbalanced genres, inconsistent annotations), typological findings based on annotated corpora with large amounts of authentic linguistic material (rather than those based on sample sentences) genuinely advance the field of typology and bring it closer to actual usage data.

Moreover, initial studies using annotated data have been carried out to re-examine some word order universals. Some of these studies have challenged a particular Greenbergian universal theoretically or computationally (e.g., Her 2017; Merlo and Ouwayda 2018), such as Universal 20, viz., “When any or all of the items (demonstrative, numeral, and descriptive adjective) precede the noun, they are always found in that order. If they follow, the order is either the same or its exact opposite” (Greenberg 1963: 87). Other studies have examined several Greenbergian universals under specific frameworks (e.g., Gerdes et al. 2019; Sharma et al. 2019). These existing studies have provided new insights into modern linguistic typology, but it is still necessary to emphasize the importance of approaching word order typology through frequency and probability.

Quantitative approaches have also played an essential role in typological studies. Notably, Greenberg (1963) did not use statistical methods, but frequently uses expressions in his statements such as “almost always”, “with overwhelmingly greater than chance frequency”, “usually”, and so forth. Even some early works emphasized this probabilistic tradition (e.g., Greenberg 1960). More recently, statistical methods, such as multiple regression, have been applied to testing language universals (e.g., Bentz and Winter 2013; Bickel 2013; Cysouw 2010; Jaeger et al. 2011; Östling 2015; Sinnemäki 2010). These studies emphasize the statistical nature of linguistic typology.

The current study implements annotated corpora and quantitative approaches to Greenbergian universals. It focuses on the first five universals (Greenberg 1963: 76–79) as a preliminary step toward verifying these universals. Specifically, the universals under investigation are:

Universal 1. In declarative sentences with nominal subject and object, the dominant order is almost always one in which the subject precedes the object.

Universal 2. In languages with prepositions, the genitive almost always follows the governing noun, while in languages with postpositions it almost always precedes.

Universal 3. Languages with dominant VSO order are always prepositional.

Universal 4. With overwhelmingly greater than chance frequency, languages with normal SOV order are postpositional.

Universal 5. If a language has dominant SOV order and the genitive follows the governing noun, then the adjective likewise follows the noun.

The noteworthy point is that the Greenbergian universals are usually based on the term *dominant order*, which occurs 13 times in 45 universals (in expressions such as “dominant VSO order”, “dominant VSO”, “dominant order VSO”, etc.). However, this phrase was not quantitatively defined by Greenberg (1963). Although some subsequent studies (e.g., Croft 2003: 60; Whaley 1996: 100–106) have proposed some criteria for dominant order, nearly all of them have been based on native speaker intuition or have considered what is more frequent in language use. One more quantitative definition for dominant order has been proposed in the *WALS* (see Section 2.2.4; Dryer 2005: 371; Dryer 2013a).

Here, we adopt the definition in the *WALS* to determine the discrete dominant orders of word order relations that are retrieved from annotated corpora. Then, we compare these dominant order results with those in the *WALS* to examine the viability of this approach. Finally, we re-examine the selected Greenbergian universals in two ways, first by classifying word order into discrete categories as in the *WALS* and then by approaching word order as continuous data from annotated corpora. We construct mixed-effects models to evaluate the universals under discussion.

Two research questions are relevant here:

- (1) When word order is classified into discrete categories, can the dominant orders extracted from annotated corpora validate the Greenbergian word order universals?
- (2) When word order is approached as continuous frequency data, is it possible to predict the Greenbergian universals?

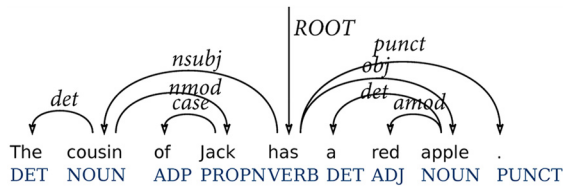


Figure 1: UD representation of the sentence *The cousin of Jack has a red apple.*

These two questions aim to test whether the frequency data extracted from annotated corpora can validate well-known typological universals from two perspectives.

2 Materials and methods

2.1 Materials

We adopted the Universal Dependencies (UD) treebank version 2.5 (Zeman et al. 2019)¹ as the source for our corpus linguistic material. It is a database of 157 annotated corpora that aims to maximize crosslinguistic parallelism by consistently annotating different languages (Nivre 2015). One of the most significant advantages of this database for typological studies is its consistent and specific annotations, so different languages can be easily compared and interpreted. Figure 1 is an example sentence based on the UD annotation scheme.

In Figure 1, the sentence is segmented into eight words. The labeled arrows between heads/governors and dependents form different dependency types, which can be used to reveal the syntactic relations between different linguistic elements within a sentence (Heringer 1993; Hudson 1995; Tesnière 1959). For example, the word *cousin* is the nominal subject of the verb *has*, and its dependency type in the UD annotation is *nsubj*.² The word *apple* is the nominal object of the verb *has*, and its dependency type is *obj*. Thus, the word order of this sentence is subject-verb-object (SVO). In addition, the dependency type between *cousin* and *Jack* is *nmod* (nominal modifier),³ and the *nmod* relation points from *cousin* to *Jack*. Hence, this possessive noun phrase has a noun-genitive (NGen) order. In this way, the dependency relations in the UD treebanks can be used for analyzing the word order of linguistic elements within sentences.

2.2 Methods

2.2.1 Selecting annotated corpora

There are 157 annotated corpora of 90 languages in UD 2.5. For languages with more than one corpus, we combined them into a single corpus, obtaining 90 corpora of 90 languages altogether. Moreover, we excluded corpora when their frequencies of any ternary or binary combinations were smaller than 20 to keep a consistent

¹ UD 2.5 is available at <http://hdl.handle.net/11234/1-3105>. An alternative database is SUD (Gerdes et al. 2018; <https://gitlab.inria.fr/grew/SUD>). One of the most important differences between these is that UD 2.5 has more detailed and specific dependency relation annotations than SUD (Yan and Liu 2022), which makes UD 2.5 potentially more applicable for the current study.

² We departed from Greenberg (1963), which focused only on noun subjects and noun objects, by taking all nominal subjects and objects (including pronouns) into account to examine the word order features in a broader sense.

³ In this paper, we used the dependency relation *nmod* to identify possessive relations between the genitive and its head noun (see Appendix A for details). Note that the dependency relation *nmod* codes for any type of relation that a noun may have to its governing noun, including, for instance, *a book on the table* and *a leg of the table*. This may lead to some false positives. But since this is a design feature of the current UD tagging (see <https://universaldependencies.org/u/dep/nmod.html>), it would be extremely time-consuming to check through all treebanks of all 74 languages manually (also, as Sinnemäki and Haakana [2020] found, computer-assisted algorithms perform poorly in detecting possessive constructions from the UD treebanks that are not marked morphologically). Therefore, the present study used the *nmod* relation as a compromise to represent genitive-noun constructions.

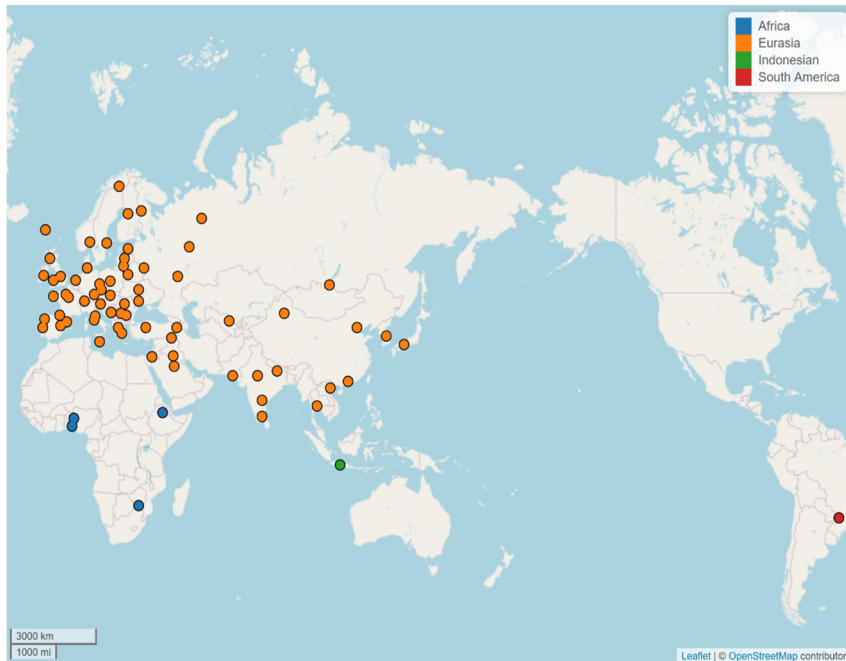


Figure 2: Macro-areas represented by the 74 languages in the selected corpora. The areal distribution was based on Glottolog 4.4 (Hammarström et al. 2021) and generated by the package *lingtypology* (version 1.1.5; Moroz 2017).

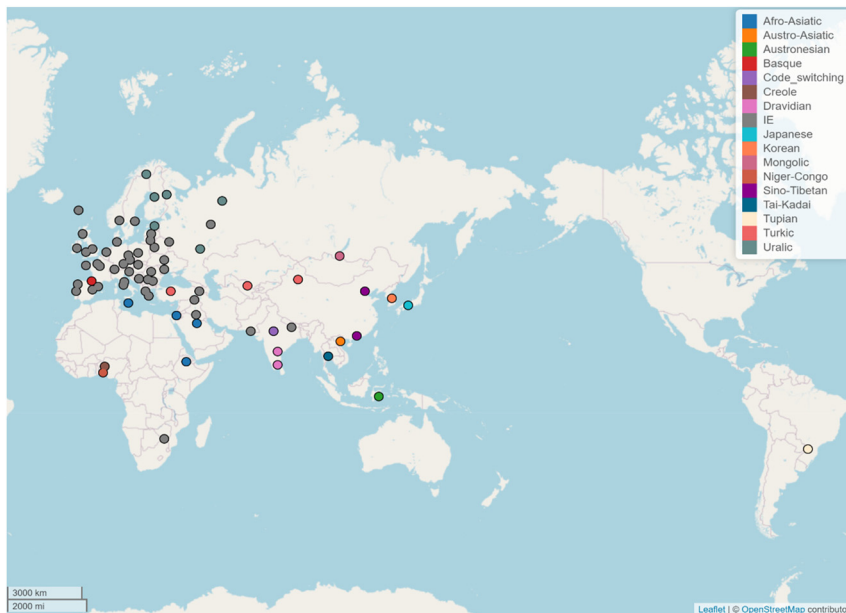


Figure 3: Language families represented by the 74 languages in the selected corpora. The areal distribution was based on Glottolog 4.4 (Hammarström et al. 2021) and generated by the package *lingtypology* (version 1.1.5; Moroz 2017).

number of languages for all our analyses (cf. Levshina 2019: 542). After excluding 16 languages on this basis, 74 languages remain.

According to Glottolog 4.4 (Hammarström et al. 2021), these 74 languages belong to four major macro-areas (Figure 2), 17 major language families (Figure 3), and 32 language genera (see Appendices B and C).

2.2.2 Extracting word orders

As Nichols (2007: 231) puts it, one essential feature of typology is its “concern with observable phenomena that pattern interestingly in the world’s languages (in their frequency, their interaction with other parts of grammar,

their geography, their history, etc.)”. To extract such patterns from UD 2.5, we followed Bonfante et al. (2018), Courtin (2018: 36–40), and Yan and Liu (2021) to retrieve the frequencies of six ternary linear relations (i.e., VSO, SVO, SOV, OSV, OVS, VOS) and eight binary relations (i.e., VO, OV, NGen, GenN, NAdp, AdpN, NAdj, AdjN) for the investigation of the first five Greenbergian universals (see Appendix A for details).

2.2.3 Calculating word order probabilities

Nearly all levels of linguistic representations show probabilistic properties (Bod et al. 2003). In this study, the occurrence of one specific linear order of a ternary or binary relation extracted from the corpora, that is, one value of a word order parameter, was taken as the numerator. The frequency of all possible values of a word order parameter was taken as the denominator. Thus, a specific order’s probability, or the relative frequency (Evans and Rosenthal 2004), can be calculated. Based on the probability, we further our discussion in Section 3 by viewing word order as discrete versus continuous data.

2.2.4 Determining dominant word orders

Following Dryer (2005: 371; 2013a), we determined the dominant word order for each word order parameter in all 74 languages. For binary relations, when one order was more than twice as common as the other, the more frequent one was classified as the dominant one. Otherwise, the language was classified as lacking a dominant order for the binary relation. For ternary relations, when the most frequent order was more than twice as common as the second most frequent order, we classified the most frequent order as dominant. Otherwise, the language was classified as lacking dominant order for the ternary relation.

2.2.5 Predicting word order parameters

We used mixed-effects models to predict the relationship between the word order parameters of each Greenbergian universal. Technically, based on the R software (R Core Team 2021), we employed the packages lme4 (version 1.1.23; Bates et al. 2015) and lmerTest (version 3.1.3; Kuznetsova et al. 2017) to perform the mixed-effects analyses.

3 Results and discussion

This section first compares the dominant word orders based on corpora with those in the *WALS*. If the null hypothesis of no difference between dominant word orders based on the annotated corpora and those retrieved from the *WALS* cannot be rejected, we then test whether the corpus data uphold the Greenbergian universals.

In the *WALS*, six chapters are related to Greenberg’s first five word order universals (Dryer 2013b, 2013c, 2013d, 2013e, 2013f, 2013g). We compared the dominant word orders extracted from the 74 corpora (Section 2.2.4) and those in the *WALS* database.

In Table 1, the approximate chi-square tests indicate that the null hypothesis of no difference between dominant word orders based on the corpora and those retrieved from the *WALS* could not be rejected ($p > 0.05$). The effect sizes were also weak (Cramer’s $V < 0.3$).

Here, we take the dominant word orders of subject, object, and verb (the second row of Table 1) as an example to better demonstrate the consistency between these two data sets.

As shown in Table 2, 51 languages matched between the UD treebanks and the *WALS*. Among these 51 languages, the most frequent dominant word order was SVO: this was the dominant order in 31 languages in both the UD treebanks and the *WALS*. The second most frequent dominant order was SOV: found in eight languages in the UD treebanks and 12 in the *WALS*. Then, VSO dominant order was not so frequent: only in four languages in both data sets. Finally, eight languages lacked dominant order based on the UD treebanks and four in the *WALS*.

Table 1: Permutation tests to assess the independence of the dominant word orders based on the UD treebanks and those in the *WALS*.

Word order	χ^2	<i>p</i> value	Cramer's <i>V</i>
Order of subject, object, and verb	2.13	0.56	0.14
Order of subject and verb	0.24	0.89	0.05
Order of object and verb	3.55	0.20	0.19
Order of adposition and noun phrase	3.13	0.33	0.18
Order of genitive and noun	4.87	0.09	0.23
Order of adjective and noun	0.00	1.00	0.00

Note: Permutation tests were adopted for the hypothesis testing since they are effective when the assumptions of parametric statistics are not met or when the sample size is too small. The *p* values were obtained from the observed statistic and null distribution based on 9,999 replications.

Table 2: Contingency table of the dominant word orders of subject, object, and verb based on the UD treebanks and those in the *WALS*.

Word order	UD treebanks	<i>WALS</i>
SVO	31	31
SOV	8	12
VSO	4	4
Lacking a dominant order	8	4

In summary, the only difference between these data sets seems to be that four languages that were classified as SOV in the *WALS* were classified as lacking a dominant order in the UD treebanks.

The chi-square independence test indicates that the results of dominant order for subject, object, and verb based on the UD treebanks are not significantly different from those based on the *WALS* ($p = 0.56$; i.e., $p > 0.05$). A similar result was also found for the other five sets of word orders in Table 1. Hence, these results suggest that there is, by and large, a consistency between the dominant orders extracted from the UD treebanks and those based on the *WALS*. This consistency paves the way for the following validation of the Greenbergian universals based on the word order information extracted from the UD treebanks.

3.1 Viewing word order as a discrete variable

In Section 3.1, we view word order as a discrete category and examine Greenberg's first five universals one by one. Universal 1 concerns the word order of subject and object in declarative sentences.⁴ The dominant word orders of subject and object retrieved from the UD treebanks showed that the word order in 71 out of 74 treebanks was SO dominant, and in the other three, there was no dominant word order (see Appendix D). Based on the data, we generated an intercept-only mixed-effects model with the order of subject and object as the dependent variable, and language genus and geographical area as the random effects (formula: $S_O \sim 1 + (1 | \text{Genus}) + (1 | \text{Area})$).⁵ The results ($\beta = 8.89$, $SE = 3.14$, $z = 2.83$, $p < 0.01$) show that languages are likely to be SO dominant ($\beta = 8.89$), and that

4 Universal 1 is based on declarative clauses; hence it is essential to ensure this presupposition. In English, for example, Biber et al. (1999) pointed out that there exist four clause types: declarative, imperative, interrogative, and exclamative. Since question and exclamation marks are essential punctuation in imperative, interrogative, and exclamative clauses, we removed all sentences with these punctuation marks.

5 In an early draft of this manuscript, we included only language genera as the random effect (Baayen et al. 2008). The reason that we did not take geographical information into account is that we assumed the geographical information of the languages under investigation is not so suitable for constructing a model since a random effect is supposed to have at least five levels to accurately estimate within-group variations (Gelman and Hill 2006; Harrison et al. 2018), while the 74 languages cover only four macro-areas in Glottolog 4.4 (see Figure 2). Here, we would like to thank Dr. Kaius Sinnemäki for suggesting that we could employ the geographical areas retrieved

this tendency is statistically significant ($p < 0.01$). In other words, the model verifies Universal 1. This result is also consistent with Bickel (2013), which found a universal pressure or a directional family bias for languages to place agents before patients (A-before-P orders) under the assumption of the Family Bias Theory.

The second universal is bidirectional. To test whether the order of adposition and noun can predict the order of noun and genitive, we constructed two models by using language genus and geographical area as random effects. The results show that prepositional languages are prone to be NGen dominant ($\beta = -19.83$, $SE = 5.90$, $z = -3.36$, $p < 0.001$) and postpositional languages tend to be GenN dominant ($\beta = -22.17$, $SE = 6.70$, $z = -3.31$, $p < 0.001$). Therefore, the mixed-effects models confirm Universal 2.

We focused on Universal 3 and Universal 4 by generating mixed-effects models with the word order of verb and object and that of adposition and noun as the independent and dependent variables, respectively. The reason we did not adopt the six-way division of SVO, SOV, VSO, OVS, VOS, and OSV is that earlier studies (Dryer 1992; Lehmann 1973; Vennemann 1973) have suggested that the two-way classification of VO and OV is more compatible with the other binary word orders. Hence, we simplified these ternary relations into binary relations in both Section 3.1 and Section 3.2. The models show that languages with VO dominant order tend to be prepositional ($\beta = -22.85$, $SE = 6.03$, $z = -3.79$, $p < 0.001$), and OV dominant languages are likely to be postpositional ($\beta = -23.33$, $SE = 5.79$, $z = -4.03$, $p < 0.001$). The results are thus consistent with Universal 3 and Universal 4.

Finally, for Universal 5, we constructed two mixed-effects models to predict the relationship between the dependent variable (the word order of adjective and noun) and the independent variables (the word order of verb and object and that of genitive and noun). In one model, we included the interaction between the two independent variables (model 1); in the other, we excluded it (model 2). According to the Akaike information criterion (AIC: 62.91 [model 1] < 66.78 [model 2]),⁶ we found that the model with the interaction (model 1) was better supported. The interaction between the two independent variables (NGen:OV, in lme4 syntax) shows that NGen and OV dominant languages are not likely to be NAdj dominant ($\beta = -17.04$, $SE = 8.12$, $z = -2.10$, $p = 0.04$, $p < 0.05$) since a negative slope ($\beta = -17.04$) means that the interaction does not favor the baseline (Noun-Adjective dominant order) of the dependent variable. This result thus indicates that these two independent variables cannot be used to predict the dependent variable, thus providing evidence against Greenberg's Universal 5.

To recap, viewing word order as a discrete variable shows that the first four universals are verified by the corpus data, while the fifth one is not. These results are in line with earlier typological research that draws the data from reference grammars (e.g., Dryer 1989: 279; 1992). As an alternative to treating word order as a discrete variable, in Section 3.2 we treat the word orders of different languages as continuous variables to test the universals and explore other possible solutions to characterize the word orders of natural languages.

3.2 Viewing word order as a continuous variable

As mentioned above, it is inaccurate and perhaps even misleading to simply categorize a specific language into a dominant order type. All languages, to some extent, are scattered on a continuum of word order preference (Filipović 2013; Liu 2010). This section addresses this issue by modeling the predictors and response variables as proportions when constructing the models. Language genus and geographical area are again used as random effects.

First, we focused on Universal 1 by constructing an intercept-only mixed-effects model with SO proportion as the dependent variable (formula: $SO \sim 1 + (1 | \text{Genus}) + (1 | \text{Area})$). The intercept of the model ($\beta = 0.85$) was positive and significant ($p < 0.001$), meaning that languages are likely to be SO dominant ($\beta = 0.85$, $SE = 0.04$, $t = 20.13$, $p < 0.001$). This result confirms Universal 1.

from the AUTOTYP database (version 1.1.0; Bickel et al. 2022) to take not only genealogical affiliation but also geographical affiliation into consideration.

⁶ AIC is a model comparison method that considers not only a model's statistical fit but also the number of parameters. Generally, models with smaller AIC values are preferred in model construction (Kabacoff 2011).

Then we turned to Universal 2. It is bidirectional; hence we generated two mixed-effects models to predict their interrelationship. The first one was fitted with the order of NGen and the order of AdpN as the dependent and independent variables, respectively (formula: $\text{NGen} \sim \text{AdpN} + (1 | \text{Genus}) + (1 | \text{Area})$). The model shows that as the probabilities of AdpN increase, the probabilities of NGen increase ($\beta = 0.75$, $\text{SE} = 0.09$, $t = 8.07$, $p < 0.001$), since a positive slope ($\beta = 0.75$) means that two variables are positively related. The result confirms that prepositional languages tend to have genitives following their governing nouns. The second model was generated to predict the order of GenN from the order of NAdp (formula: $\text{GenN} \sim \text{NAdp} + (1 | \text{Genus}) + (1 | \text{Area})$). The result also indicates that as the probabilities of NAdp increase, the probabilities of GenN increase ($\beta = 0.75$, $\text{SE} = 0.09$, $t = 8.07$, $p < 0.001$), which is consistent with Universal 2. Therefore, the two models verify Greenberg's Universal 2.

Next, we tested Universal 3 and 4. Both universals can be quantified as modeling the order of adposition and noun as a function of the order of verb and object. Hence, we generated a mixed-effects model with the proportions of VO and AdpN as the independent and dependent variables, respectively. The model shows that the proportion of AdpN increases with the increase of VO probabilities ($\beta = 0.06$, $\text{SE} = 0.03$, $t = 2.14$, $p = 0.04$, $p < 0.05$). The result concurs with Universal 3. Similarly, with the increase of OV proportions, the proportions of NAdp also increase ($\beta = 0.06$, $\text{SE} = 0.03$, $t = 2.14$, $p = 0.04$, $p < 0.05$), aligning with Universal 4.

Finally, we turned to Universal 5. Two predictors and one response variable are involved in this universal; hence we generated two mixed-effects models. One model predicts NAdj from the interaction of the OV and NGen proportions (formula: $\text{NAdj} \sim \text{OV} * \text{NGen} + (1 | \text{Genus}) + (1 | \text{Area})$). The other model predicts NAdj from OV and NGen proportions (formula: $\text{NAdj} \sim \text{OV} + \text{NGen} + (1 | \text{Genus}) + (1 | \text{Area})$). The results show that the AIC of the second model (34.75) is smaller than that of the first model (36.36; though the difference is less than two units). In addition, the interaction (OV:NGen) of the two independent variables in the first model is insignificant ($\beta = -0.20$, $\text{SE} = 0.31$, $t = -0.64$, $p = 0.52$, $p > 0.05$). Hence, there was little support for the interaction term. In other words, the second model without the interaction between OV and NGen proportions (formula: $\text{NAdj} \sim \text{OV} + \text{NGen} + (1 | \text{Genus}) + (1 | \text{Area})$) is better than the first one. In the second model, the slope of the OV proportions is negative but insignificant ($\beta = -0.13$, $\text{SE} = 0.13$, $t = -1.03$, $p = 0.31$, $p > 0.05$), and that of the NGen is positive but insignificant ($\beta = 0.29$, $\text{SE} = 0.15$, $t = 1.88$, $p = 0.07$, $p > 0.05$). The results indicate that as the OV proportions decrease and the NGen proportions increase, the probabilities of NAdj increase, meaning that the direction of the NGen and OV proportions and the direction of the NAdj proportion are not the same. More importantly, these correlations are not significant. Hence, these results contradict Universal 5, suggesting that it might be difficult to predict NAdj word order from the order of NGen and OV.

To summarize, we revisited Greenberg's first five basic word order universals by using continuous variables. The results were similar to what we found when we viewed word order as a discrete variable. The findings illustrate the practicability of adopting continuous probability-based data in validating word order universals.

4 Further discussion

Regardless of whether word order is approached as a discrete or a continuous variable, the results seem to falsify the fifth Greenbergian universal. Consequently, we examined the language samples in the UD treebanks in more detail. According to the word order probabilities retrieved from the treebanks, only three languages have the dominant SOV and NGen orders required by Universal 5: Persian, Afrikaans, and Swiss German (see Appendix D). This low frequency is consistent with Dryer's (2007: 111) statement that, in SOV languages, genitives usually precede their governing nouns (GenN order), so the number of languages satisfying the criteria for Universal 5 is inherently small.

Moreover, based on the UD treebanks of these three languages, only Persian is NAdj dominant. The other two generally have adjectives preceding their modifying nouns. According to Greenberg (1963: 109), languages with SOV&AdpN&NGen&AdjN dominant orders were not found in his samples. However, our results retrieved from

the UD treebanks show that Afrikaans and Swiss German might violate this universal since they are SOV&AdpN&NGen&AdjN dominant.⁷ So, what are the characteristics of these languages?

To better observe the word order features of these two languages, we also discuss Dutch and German here since they are the languages that would be expected to share similar characteristics with Afrikaans and Swiss German in the UD database. In Appendix F, we present these languages' dominant orders based on the WALS and the UD treebanks (with word order probabilities).

The results show that Afrikaans and Swiss German demonstrate stronger inclinations toward SOV than Dutch and German. This tendency of Afrikaans is consistent with some previous statements that the basic order of Afrikaans is SOV + V2 (e.g., den Besten 2002: 4) and explains why some studies have classified Afrikaans as SOV dominant directly (e.g., Yadav et al. 2020: 7). It is also the case that Swiss German can be regarded as an SOV dominant language (e.g., Schönenberger 2014; van Riemsdijk 2012).

Hence, this discussion suggests that the set of languages that are SOV&NGen&AdjN dominant is small and internally heterogeneous. In other words, if a language has dominant SOV order and the genitive follows the governing noun, it is not so likely to have adjectives following the nouns.

5 Concluding remarks

This study quantitatively revisits the first five Greenbergian word order universals based on the word order information extracted from 74 annotated corpora. The results show that Greenberg's first four basic word order universals are well verified, while the fifth is falsified by the modeling and more detailed inspection of some sample languages. Previously, Sharma et al. (2019) examined several Greenbergian universals based on network approaches, but they could not deduce Universal 3 from the network parameters, and Universal 5 was refuted. Our results quantitatively validate Universal 3 and re-confirm the refutation of Universal 5. Moreover, the frequencies and probabilities data are more concrete, specific, and consistent with the nature of the Greenbergian statistical universals than the abstract network parameters.

As for the counterexamples found, we agree that “the universal is a tendency, rather than an absolute” (Comrie 1989: 158), and “methods other than typological statistics are necessary to establish absolute properties of human language” (Piantadosi and Gibson 2014: 736). The findings suggest that annotated corpora can be well applied in research on word order typology. Meanwhile, the current study demonstrates the importance of carefully adopting the most basic principles that are taken for granted by the typological community. Also, it indicates that further investigation is needed into the renowned universals and provides insights for future related studies.

As a preliminary attempt to verify the Greenbergian universals with annotated corpora, this study has several limitations. First, it only focuses on the first five universals of the basic order typology. More typological universals await further investigations with various materials and approaches. Second, the sizes and genres of the language corpora vary, so it is unclear whether the results represent language use more generally or are limited to what the corpora represent. Third, the language samples are not well balanced in terms of genera and geographical locations, though we adopted mixed-effects modeling to counteract the effects of genealogical and geographical affiliations.

In short, this study, on the one hand, demonstrates the feasibility of adopting frequencies and probabilities in validating Greenbergian universals from both discrete and continuous perspectives, and on the other hand, calls for more future explorations into various typological universals and linguistic laws based on more balanced corpora with more evenly distributed language samples.

⁷ Some studies also present examples that violate the Greenbergian Universal 5. For instance, Dryer (2007: 111) pointed out that Tigre (Raz 1983), an SOV&NGen&AdjN dominant language spoken in Eritrea, contradicts Universal 5. Moreover, Singh and Singh (2021: 168f) found that Moyon, similar to Manipuri, is an OV&NGen dominant language, but it demonstrates both possibilities of AdjN and NAdj word orders without any preference.

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